SETLLA PLATFORM ARCHITECTURE & SYSTEM OVERVIEW

Comprehensive Platform Specification

Verified Rentals, Jobs, Partner Ad Network & Super Admin Command

Setlla is a high-trust, dual-sided real estate and employment marketplace designed for Nigeria (with deep density across Delta State: Asaba, Warri, Abraka, Ughelli, Sapele) and scaling nationwide. The platform eliminates ghost listings, fake agents, and job recruitment scams through automated freshness lifecycles, verified profiles, and escrow-backed inspections.

1 Core Platform Pillars (The 4 Ecosystems)

Pillar 1: Verified Property Marketplace & 72-Hour Freshness Guarantee

Automated 48-Hour Availability Checks prompt agents via 1-tap freshness confirmations. Unconfirmed properties safely auto-delist after 72 hours to guarantee zero ghost listings. Listings marked "Rented Out" remain visible for 24h as social proof before being purged. Includes 1-click direct WhatsApp routing, direct phone calling, and 24-hour escrow inspection pass protection.

Anti-Ghost Engine | 72h Freshness | 1-Click WhatsApp | Inspection Escrow

Pillar 2: Setlla Jobs & Recruiter Recruitment Portal

A dual recruitment space connecting job seekers with verified employers across hospitality, logistics, tech, healthcare, and retail. Employers post vacancies, review resumes, and manage candidates in real-time. Verified Job Agents have specialized tools to represent and recruit for multiple commercial employers simultaneously.

Job Seekers | Employer Dashboard | Resume Ingestion | Job Agents

Pillar 3: Sponsored Ad Network & Multi-Channel Campaigns

Commercial promotion ecosystem with 4 high-visibility placements: Feed Stories Tray, Pinned Desktop Sidebar, In-Feed Video Billboards, and Jobs Feed Banners. Offers 3 ingestion paths: Self-Service ("Post Ad"), Self-Service Partner Portal with wallet and campaign tiers (Starter 7d, Growth 30d, Corporate 90d), and a Programmatic REST

Stories Tray | Desktop Sidebar | In-Feed Video | Programmatic API | Paystack Gateways

Pillar 4: Super Admin Command Center & Security Watchdog

Direct operational command over platform governance. Features live KPI counters, financial tracking, user directory management, listings & jobs oversight, and advert campaign approvals/declines with recorded feedback. Built with multi-layer authorization: 8-digit OTP via Brevo, Elvis Super Admin profile lock, 60-minute inactivity auto-logout, and cold-resume tab protection.

Brevo 8-Digit OTP | Elvis Profile Lock | 60-Min Idle Logout | Audit Trail Logging

2 Technology Stack Breakdown

Frontend Framework & Architecture

React 19 + TypeScript + Vite. Ultra-fast, type-safe compilation with a responsive split-column layout. Fully styled using Tailwind CSS with Setlla's custom warm earthen palette (#000004, #342A21, #FCF9F5, #F95700).

Database, Realtime & Cloud Storage (Supabase)

PostgreSQL relational database via Supabase. Realtime WebSocket subscriptions (setlla_realtime_adverts, setlla_realtime_listings, setlla_realtime_bookings) sync state across all devices instantly with zero manual refresh required.

Edge Media Delivery & Video Streaming (Cloudinary)

Direct unsigned edge CDN uploads for photos and video walkthroughs. Automatically optimizes format (f_auto) and quality (q_auto), and extracts video poster frames (so_0). Paired with AdvertVideoPlayer for continuous, unblocked HTML5 autoplay.

Payment Infrastructure (Paystack)

Live Paystack NGN gateway integration. Handles card payments, bank transfers, and USSD. Automates transaction reference verification, advert sponsorship activation, and inspection fee escrows.

Transactional Email Engine (Brevo / Sendinblue)

High-deliverability transactional email delivery. Generates branded HTML emails containing 8-digit authorization OTP codes for Super Admin access and irreversible deletion notifications.

Progressive Web App (PWA) & Offline Cache

Modern Service Worker (public/sw.js) caching, Web App Manifest, App Badging API, Web Push Notifications, and standalone device installation detection (beforeinstallprompt, getInstalledRelatedApps).

3 Security, Anti-Fraud & Data Lifecycle

Multi-Tier Anti-Ghost Tombstone Architecture

Deleted adverts and listings are synchronized via cloud tombstones (deleted_adverts and deleted_listings tables). When a listing or advert is permanently deleted by a user or admin, its tombstone propagates across devices so it never resurrects in cached local storage.

Cloud Tombstones | Zero Resurrection | Realtime Pruning

Super Admin 60-Minute Idle Watchdog

Monitors active user interactions (pointerdown, keydown, scroll, touchstart) with a 10-second throttling buffer. If an administrator is idle for 55 minutes, a non-intrusive warning appears; at 60 minutes, the session is cleared and the portal re-locks.

Interaction Watchdog | Cold Resume Detection | Auto-Purge

4 Technical Architecture Matrix

Component	Technology / Service	Primary Role & Guarantees
Frontend UI	React 19, Vite, Tailwind CSS	SPA client with responsive mobile-first UX
Database	Supabase (PostgreSQL)	ACID transactions, relational schema & RLS
Realtime Sync	Supabase Realtime Channels	Instant multi-device broadcast of updates
Media CDN	Cloudinary Unsigned API	Edge video streaming, q_auto & poster frames
Local Storage	IndexedDB + localStorage	Offline video/photo cache & device registry
Payment Gateway	Paystack Live API	NGN card, transfer & USSD instant checkout
Transactional Mail	Brevo (Sendinblue) REST API	Super Admin 8-digit OTP & account notices
PWA & Mobile	Service Worker + Manifest	App badging, offline fallbacks & installable app